INKLOOM AI HACKATHON • GRAND FINALE CHAMPIONSHIP

BRANDGUARD AI

Autonomous Pre-Publication Brand Governance, Multi-Jurisdiction Regulatory Enforcement & Cryptographic Stage-Gate Clearance

Team Synapse: Aman Sharma (Sole Administrator & Backend Lead) • Janhavi Kanojia (Frontend & UI/UX Developer)

Production Deployment: https://brandgovernanceai.onrender.com • **GitHub:** github.com/CodingWithAman1/BrandGovernanceAI

Core Architecture: Dual Deterministic-Neural Hybrid (5 Tone Polarity Spectrums, SQL Truth Bank, Pillow K-Means WCAG 2.1 AA, Google Gemini 3 Flash)

VERIFICATION MATRIX	SOLE ADMIN LOCK	PUBLIC AUDIT LEDGER	CI/CD INTEGRATIONS
24 / 24 Automated Tests (100% Pass)	aman@synapse.ai (Sole Administrator)	SHA-256 Zero-Login Verifier (/verify)	GitHub Actions Gate (/api/v1/gate/check)

THE PROBLEM STATEMENT & MARKET OPPORTUNITY

High-Velocity Generative Content vs. Strict Regulatory Compliance

1. FTC Section 5 & AI Comply

- FTC 15 U.S.C. § 45 strictly prohibits deceptive marketing.
- *Operation AI Comply* penalizes ungrounded '#1' claims and unverified speed superlatives.
- **Penalty:** Up to $51,744/day per violation and civil consent decrees.

2. SEC AI-Washing Rule

- SEC Rule 10b-5 targets deceptive forward-looking AI statements.
- Companies claiming 'autonomous intelligence' or '100% zero risk' face securities enforcement.
- **Penalty:** Executive bans, shareholder lawsuits, and mandatory fraud disclosures.

3. EU AI Act Art. 50/52 Mandates

- Mandates transparent synthetic content watermarking and disclosure.
- Requires auditable human oversight (HITL) for high-impact promotional copy.
- **Penalty:** Fines up to €35M or 7% of annual worldwide revenue.

4. ADA Title III & WCAG 2.1 AA Contrast

- Federal courts require digital marketing to satisfy WCAG 2.1 contrast (4.5:1 ratio).
- Over 4,000 accessibility lawsuits filed in 2023 targeting low-contrast banners.
- **Penalty:** Statutory attorney fees, mandatory settlement decrees, and injunctions.

ROLE-BASED GOVERNANCE ARCHITECTURE

Three Mathematically Isolated Portals with Strict Sole Admin Lock

1. Client / Creator Workspace

Role: CREATOR • /portal/client
• Pre-Publish Compliance Studio with multi-modal scanning.
• 1-Click suggested fixes with live score recalculation.
• AI Deep Semantic Harmonization via Google Gemini 3 Flash.
• Business exception rationale submission.
• Personal submission tracking & active certificates.

2. Reviewer Desk & Legal Queue

Role: REVIEWER / MANAGER • /portal/reviewer
• Centralized Stage-Gate review queue with SLA timers.
• Side-by-side red/green diff inspector with inline flag fixes.
• Full visibility into author exception rationales & citations.
• 1-Click Stage-Gate clearance (Approve, Request Changes, Reject).
• Automatic cryptographic SHA-256 PDF certificate compilation.

3. Executive Admin Command Plane

Role: ADMIN • Strictly locked to **Aman Sharma**.
• Brand Rulebook CRUD with regex and severity controls.
• SQL Truth Bank registry with evidence citations.
• AI Ambiguity Resolver with creator dispute telemetry.
• Clarity Index Analytics with Brand Drift Time-Travel.
• Enterprise user activation/deactivation controls.

EVALUATOR & JUDGE DEMO MODE: 1-CLICK PERSONA SWITCHER IN NAVBAR
To provide an effortless evaluation experience for hackathon judges, BrandGuard AI embeds an agile Persona Quick-Switcher directly into the header. Evaluators can switch seamlessly between Priya (Client Creator), Elena (Legal Reviewer), Marcus (Brand Director), and Aman Sharma (Sole Administrator) with full session re-authentication in 0.04s.

THE DETERMINISTIC & NEURAL ENGINE ARCHITECTURE

Four Deterministic Compliance Engines Combined with Google Gemini 3 Flash

Engine 1: Tone Calibrator

- 5 Voice Polarity Vectors & lexical filters.
- Scans for 12 robotic AI filler clichés (*delve into*, *rich tapestry*, *game-changer*).
- Flags combative attacks & ungrounded superlatives.
- Calculates Flesch Reading Ease and Formality Index.

Engine 2: SQL Truth Bank Fact-Checker

- Cross-references claims against the ApprovedFact repository.
- Flags unverified superlatives (*#1 in the world*, *100% risk-free*).
- Docks -20 points per ungrounded regulatory claim.
- Enforces FTC Act Section 5 substantiation standards.

Engine 3: Pillow K-Means WCAG Inspector

- Extracts dominant palette colors using K-Means clustering.
- Enforces WCAG 2.1 AA luminance formula: Cr = (L1 + 0.05) / (L2 + 0.05).
- Strict 4.5:1 minimum contrast threshold for text copy.
- Prevents ADA Title III digital accessibility lawsuits.

Engine 4: Google Gemini 3 Flash Co-Pilot

- Resilient multi-model cascade (gemini-3.1-flash-lite → gemini-flash-lite-latest → Local NLP).
- Powers Deep Semantic Harmonization & Safe Harbor Disclaimers.
- Interactive slide-out governance chat drawer with live telemetry.
- Closed-loop validator prevents generative AI hallucinations.

STAGE-GATE CLEARANCE & CRYPTOGRAPHIC LEDGER

Human-in-the-Loop Workflow, Public Verifier (/verify) & SHA-256 PDF

Step 1: Multi-Modal Ingest

Creator drafts marketing text or uploads banner creatives; Laser Scanner HUD executes 4-phase audit.

Step 2: 1-Click Fix & AI

Accepts one-click suggested fixes or executes AI Deep Harmonization via Gemini 3 Flash.

Step 3: Stage-Gate Queue

Routes to Reviewer Desk; legal counsel reviews side-by-side red/green diffs and exception notes.

Step 4: Formal Clearance

Authorized reviewer executes 1-click Approval; locks copy text, approver email, and UTC timestamp.

Step 5: Public Verifier

ReportLab compiles tamper-evident PDF; registers SHA-256 digest on public /verify ledger.

PUBLIC CRYPTOGRAPHIC VERIFIER & SHA-256 AUDIT LEDGER (/verify & /verify/<cert_hash>)

- **Zero-Login External Verification:** Regulatory bodies (FTC/SEC/CMA) and enterprise customers can look up any certificate hash or scan a QR code to verify authenticity without requiring internal system access.
- **Tamper-Evident Digest Seal:** Evaluates hashlib.sha256(raw_content). Modifying even a single character in approved marketing copy immediately fails verification, eliminating post-publication tampering.
- **Official PDF Compliance Certificates:** Generates downloadable PDF certificates embedding approver signature (Aman Sharma), multi-modal compliance sub-scores, and embeddable live HTML/Markdown badges.

UI/UX DESIGN SYSTEM & LIVE MICRO-INTERACTIONS

Extraordinary Obsidian Cyberpunk Interface with 0ms Cursor & 3D WebGL

Hardware-Accelerated Cyber Cursor

- Zero-lag pointermove listener with 0ms placement.
- GPU-accelerated translate3d avoiding layout thrashing.
- Dynamic role-based colors (Cyan Client, Amber Reviewer, Crimson Admin).
- Strict cursor: text !important auto-fallback for form inputs.

Three.js 3D WebGL Holographic Core

- Local offline-ready Three.js 3D Holographic Governance Core.
- Single WebGL context architecture with RAF rendering loop.
- IntersectionObserver pauses rendering when canvas leaves viewport.
- Real-time 3D card parallax tilt with 60 FPS damping physics.

Multi-Engine Laser Scanner HUD

- Full-screen holographic modal overlay on audit submission.
- Oscillating neon laser scanline traversing down the document.
- Real-time digital progress gauge counting from 0% to 100%.
- Live terminal status ticks verifying NLP features, Truth Bank, and WCAG.

Omnichannel Live Social Previews

- Pixel-accurate live preview mockups for LinkedIn, Twitter/X, and Google SERP.
- Real-time synchronization with active Studio text input.
- Visualizes social card clipping, metadata contrast, and search snippet boundaries.
- 1-Click copy formatted clean copy into clipboard.

ROUND 3 ENTERPRISE EXPANSION

Competitor Battlecard Studio, Brand Voice DNA & CI/CD Stage-Gate

Competitor Battlecard Studio (/scanner/competitor-studio)

- Crawls competitor URLs or analyzes pasted ad copy.
- Flags unverified superlatives and AI-washing claims under FTC Section 5.
- Cross-references assertions against the SQL Truth Bank.
- Automatically synthesizes 3-pillar battlecards via Google Gemini 3 Flash: 1) AE Customer Call Pitch, 2) LinkedIn Hook, 3) Executive PR Positioning.

Brand Voice DNA Profiler (/brand-dna)

- Ingests corporate brand manifestos and founder style guides.
- Extracts quantifiable 5-axis voice polarities (Authority, Warmth, Innovation, Plainspokenness, Technical Depth).
- Synthesizes high-resonance power lexicons and banned synthetic clichés.
- 1-Click commit of generated governance rules into the active rulebook.

Enterprise CI/CD Quality Gate (/api/v1/gate/check)

- Copyable GitHub Actions workflow (brandguard-gate.yml).
- Programmatic REST API endpoint evaluates marketing copy in CI pipelines.
- Returns HTTP 422 to automatically block pull requests with scores < 85.
- Live API test playground in /integrations with cURL, Python, and Node.js.

Audit Vault & Brand Drift Time-Travel

- Dedicated Regulatory Audit Vault at /analytics/audit-vault-export.
- Exports signed JSON compliance dossiers with cryptographic seals.
- Time-travel telemetry tracks brand drift across campaign versions.
- Webhook dispatch formats for Slack and Microsoft Teams incident alerts.

VERIFICATION, DEPLOYMENT & ROADMAP

Comprehensive Test Suite, Live Cloud Deployment & Product Roadmap

24/24 Automated Tests (100% Pass Rate)

- **test_complete_suite.py (18 Tests):** Covers Tone Calibrator, Claims Checker, Visual Inspector, Clarity Engine, PDF Generator, Sole Admin Lock (Aman Sharma), Stage-Gate Lifecycle, Audit Vault JSON, and Cyber Cursor.
- **test_round3_features.py (6 Tests):** Validates Public Certificate Verifier (/verify), Developer Hub (/integrations), Competitor Battlecards, Brand Voice DNA, and CI/CD Quality Gate API (/api/v1/gate/check).
- 100% assertions satisfied with zero regressions.

Production Cloud Deployment (Render)

- **Render Cloud PaaS:** Publicly active at **https://brandgovernanceai.onrender.com**.
- **Gunicorn WSGI Server:** High-concurrency production server dynamically binding to 0.0.0.0:$PORT.
- **Automated CI/CD:** Git push to branch **main** triggers live cloud build.
- **Zero-Downtime Healthcheck:** Dedicated /healthz route for load balancers.
- **Live Repository:** github.com/CodingWithAman1/BrandGovernanceAI

FUTURE ROADMAP & EXECUTIVE CREDITS • TEAM SYNAPSE

Aman Sharma (Sole Administrator & Backend Lead) • Janhavi Kanojia (Frontend & UI/UX Lead)

- **Phase 1 (Q1 2027):** Headless Browser Crawler (Puppeteer/Playwright) to parse dynamic client-rendered React/Vue competitor web applications.
- **Phase 2 (Q2 2027):** Native Figma & Google Docs desktop plugins delivering real-time brand voice feedback inline during drafting.
- **Phase 3 (Q3 2027):** Public Blockchain Timestamp Anchoring (Hedera/Polygon) for court-admissible audit proof in regulatory due diligence.

Live Application: https://brandgovernanceai.onrender.com • **GitHub:** https://github.com/CodingWithAman1/BrandGovernanceAI.git